

Product Requirement Document (PRD)

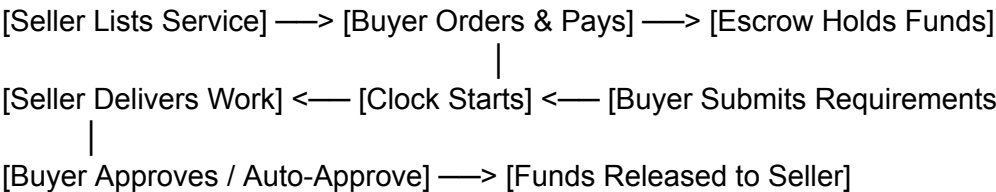
Document Overview

- **Project Title:** Service Marketplace Infrastructure Expansion
- **Author:** Product Management
- **Target Audience:** Engineering, QA, System Architects, and UI/UX Designers
- **Objective:** Upgrade the existing product-based marketplace architecture to support both **Digital Services** (files, consultations) and **Physical/On-Site Services** (local trade, logistics, styling).

1. System Overview & Core Architecture Change

The existing platform operates on an *Add to Cart* → *Checkout* → *Ship Product* logic.

The updated platform must introduce a parallel architecture: **Add Service** → **Configure Tiers** → **Checkout** → **Secure Escrow** → **Gather Requirements** → **Start Fulfillment Clock** → **Milestones/Delivery** → **Release Escrow**.



2. Epic 1: The Multi-Step "Add Service" Listing Wizard

Functional Requirements

The listing page must be converted into a multi-step form wizard that saves progress as drafts database-side on each step validation.

Step 1: Basic Metadata & Service Type Selection

- **Service Title:** Text input, max 80 characters. Pre-validated text constraint: Must begin with or naturally contain active verb phrasing (e.g., *"I will..."*).
- **Category & Subcategory Dropdowns:** Parent-child relational pickers. Selecting a category triggers a conditional UI rendering layout engine for the remaining steps.
- **Service Delivery Type Toggle (Radio Button):**

- **DIGITAL** (Online delivery of files, data, consultations)
- **PHYSICAL** (On-site tasks, local physical labor)

Step 2: Dynamic Tiered Pricing Matrix

Developers must implement a three-tiered layout matrix supporting single or triple packages (Basic, Standard, Premium).

- **Package Details:** Title (30 chars), Description (100 chars), Delivery Time (Dropdown: 1 day to 90 days), and Revisions (Dropdown: 0 to Unlimited).
- **Custom Metadata Checkboxes:** Dynamic boolean fields loaded via backend API depending on subcategory (e.g., if Category = "Web Dev", show "Source File", "Responsive Design").
- **Service Extras (Upselling Engine):** Repeatable component row letting sellers add custom line items.
 - *Fields:* Extra Description (Text), Added Price (Currency), Added Delivery Time (Dropdown days).

Step 3: Service Location & Booking (Conditional UI)

- *Visibility:* Rendered **only** if Service Delivery Type is **PHYSICAL**.
- **Geofencing Radius Set:**
 - Postal Code baseline input field.
 - Travel Radius input field (Integer) + unit dropdown (Miles/KM).
 - *Backend dependency:* Integrate Google Maps API to draw a visual boundary circle on the front-end map.
- **Availability Planner Calendar:** A grid matrix representing the 7 days of the week split into customized time slots (e.g., 9:00 AM – 5:00 PM) where the seller is available for on-site deployment.

Step 4: Media Gallery & Portfolio Slots

- **Image Upload component:** Up to 5 image slots. Constraints: WebP, JPEG, or PNG format. Max 5MB per file. Automatically optimize and compress images on the client side before Amazon S3 bucket upload.
- **Video Upload component:** 1 video slot. Constraints: MP4 format, max 50MB, max duration 60 seconds.
- **Document Upload component:** Up to 2 PDF files (useful for tech specs, qualifications, or resumes).

Step 5: The Requirements Gateway

- **Description Area:** Markdown editor supporting up to 1,200 characters explaining the service details.
- **Buyer Instructions Form Builder:** A tool for sellers to configure mandatory fields that buyers must fill after checkout.

- *Supported Input fields:* Free text box, multiple choice select, or mandatory file attachment toggle.

3. Epic 2: Checkout & The Delivery Gateway

The Requirements Gate Logic (State Machine)

Unlike products where shipping starts instantly, service orders require an explicit onboarding gate.

[Order Paid: Status = AWAITING_REQUIREMENTS]

|
(Buyer Submits Requirements)



[Order Status = IN_PROGRESS (Countdown Clock Starts)]

- **Rule:** The fulfillment delivery countdown clock **must not start**, and the order status must remain **AWAITING_REQUIREMENTS** until the buyer completes the mandatory fields designed in Step 5 of the creation wizard.
- **Fallback automated rule:** If the buyer does not submit requirements within 48 hours, an automated system alert triggers an option for the seller to cancel the order without negative impact on their rating.

The Live Order Room Feed

A dedicated workspace dashboard page created dynamically per order ID.

- **Component 1: Visual Timeline Tracking:** Displays clear milestones: Ordered → Requirements Submitted → Work Started → In Review → Complete.
- **Component 2: Real-time Communication Widget:** A WebSocket-connected text chat stream allowing text updates and attachment uploads directly inside the workspace.
- **Component 3: Countdown Timer Widget:** A real-time visual clock ticking down the hours and minutes left until the contract deadline expires. Emits a styling color shift to red when < 12 hours remain.
- **Component 4: Delivery CTA Block:** Accessible exclusively by the seller. Opens a submission box modal containing a file upload area and text description to mark the order as **DELIVERED**.

4. Epic 3: Core Financial & Database Infrastructure Upgrades

Escrow Ledger Management

The architecture must deprecate instantaneous merchant disbursement upon checkout.

- **Holding State:** When a buyer checks out, payment processors capture 100% of the funds. The funds are routed into a platform-controlled escrow holding account.
- **Release Event:** Funds are held securely until:
 1. The buyer clicks the "Accept Delivery & Release Funds" button.
 2. The order hits the **Auto-Approve Window** (system clock exceeds 3 calendar days after delivery without a dispute logged by the buyer).
- **Payout calculation formulas:**
$$\text{\text{\$}\text{\$}\text{\{Platform Commission Fee\}} = \text{\text{\{Gross Order Amount\}} \times \text{\text{\{Platform Commission Percentage\}}\text{\$}\text{\$}}$$
$$\text{\text{\$}\text{\{Seller Net Earnings\}} = \text{\text{\{Gross Order Amount\}} - \text{\text{\{Platform Commission Fee\}}\text{\$}\text{\$}}$$

Updated Database Schema Modifications (High Level)

Engineers must extend the existing schema to accommodate the following entities:

```
CREATE TABLE services (  
  id BIGINT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,  
  seller_id BIGINT REFERENCES users(id),  
  title VARCHAR(80) NOT NULL,  
  delivery_type VARCHAR(10) CHECK (delivery_type IN ('DIGITAL', 'PHYSICAL')),  
  description TEXT,  
  category_id INT,  
  status VARCHAR(15) DEFAULT 'DRAFT',  
  created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()  
);
```

```
CREATE TABLE service_packages (  
  id BIGINT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,  
  service_id BIGINT REFERENCES services(id) ON DELETE CASCADE,  
  tier_type VARCHAR(10) CHECK (tier_type IN ('BASIC', 'STANDARD', 'PREMIUM')),  
  price NUMERIC(10, 2) NOT NULL,  
  delivery_days INT NOT NULL,  
  revisions_allowed INT DEFAULT 0  
);
```

```
CREATE TABLE service_orders (  
  id BIGINT PRIMARY KEY GENERATED ALWAYS AS IDENTITY,  
  buyer_id BIGINT REFERENCES users(id),  
  service_id BIGINT REFERENCES services(id),  
  package_id BIGINT REFERENCES service_packages(id),  
  escrow_status VARCHAR(20) DEFAULT 'HELD',  
  order_status VARCHAR(30) DEFAULT 'AWAITING_REQUIREMENTS',
```

```
deadline_timestamp TIMESTAMP WITH TIME ZONE,  
created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()  
);
```

5. Non-Functional Requirements & Security Guarantees

Performance & Latency Metrics

- **Messaging System:** Chat communication logs within the Order Room must display to all active participants within an end-to-end network latency of < 300ms using persistent WebSocket loops or gRPC protocols.
- **Page Responsiveness:** Add Service layout fields must maintain smooth cross-platform parity. 100% feature availability on standard desktop viewports down to iOS Safari and Android Chrome mobile layouts.

Strict File Security Protocols

- All digital asset deliveries uploaded by users via the messaging portal or delivery modal must run through an asynchronous background anti-virus scanner hook script (e.g., ClamAV API) prior to permanent storage bucket confirmation.
- Any files flagging malicious structures or macro attachments must be immediately deleted, dropping a system alert on the user interface block.

6. Verification & Edge Case Scenarios

Test ID	Scenario Context	Operational System Expectation
TC-001	Seller tries to input a local travel radius for a DIGITAL classified service asset.	The system ui hides the physical location data modules entirely. API blocks geo payloads on digital parameters.
TC-002	Seller submits an official delivery 10 minutes past the target deadline_timestamp .	Order room appends a fixed LATE system tag to the header metadata block. Triggers automated eligibility criteria for instant buyer refunds if requested.

TC-003	Buyer submits a file deliverable review rejection, asking for modifications.	The active countdown timer pauses. Order status reverts smoothly from IN_REVIEW straight back into IN_PROGRESS state. Decrements remaining active structural revisions counter variable by 1.
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